

PAPER352 ■ R Aquarii Symbiotic Binary: FUBii with Kepler 44-Year Orbital Period

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gokshare31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF FUBii for a symbiotic nova binary with $P_{\text{orb}} = 44$ yr **Author:** Daniel T. Murphy

Abstract

R Aquarii is the nearest symbiotic nova system, consisting of a Mira Pulsating Giant and a white dwarf companion in a 44-year orbit. UQFF buoyancy-unified force FUBii ■ $-2.09 \times 10^{212} \text{ N}$ is calculated at the binary orbital radius derived from Kepler's third law: $a_{\text{orb}} = (GM_{\text{P}}/4\pi)^{1/3}$. HST 2025 near-UV imaging of the expanding jet system (launched ~10 yr ago) provides the observational anchor. The UQFF Kozima LENR coupling is relevant at the mass transfer interface between the giant's wind and the WD accretion disk.

2. Core Physics

2.1 UQFF Buoyancy-Unified Force

$$F_{\text{U_Bi_i}} \approx -2.09 \times 10^{212} \text{ N}$$

(intermediate between TDE-scale PAPER351 and AGN-scale PAPER346)

2.2 Kepler's Third Law Orbital Radius

$$a_{\text{orb}} = \left(\frac{GM_{\text{total}}}{P_{\text{orb}}^2} \right)^{1/3}$$

with $P_{\text{orb}} = 44 \text{ yr} = 1.388 \times 10^9 \text{ s}$ and $M_{\text{total}} = M_{\text{Mira}} + M_{\text{WD}} \approx 2 M_{\odot}$:

$$a_{\text{orb}} = \left(\frac{6.674 \times 10^{-11} \times 4 \times 10^{30}}{(1.388 \times 10^9)^2} \right)^{1/3} \approx 1.0 \times 10^{13} \text{ m} \approx 70 \text{ AU}$$

2.3 Mass Transfer and LENR Coupling

At the tidal mass transfer interface:

$$\dot{M}_{\text{transfer}} = \dot{M}_{\text{wind}} \cdot \left(\frac{a_{\text{orb}}}{R_{\text{Mira}}} \right)^{-2}$$

The UQFF Kozima LENR force:

$$F_{\text{Kozima}} = E_{\text{LENR}} / a_{\text{orb}}$$

where E_{LENR} is the low-energy nuclear reaction energy scale at the compressed accretion interface.

2.4 HST 2025 Jet Observation

R Aquarii's expanding jets, first resolved by HST in the 1990s and re-observed in 2025:

$$v_{\rm jet} \approx 200 \, \mathrm{km/s}$$

$$\tau_{\rm jet} \approx 10^3 \, \mathrm{yr}$$

$$L_{\rm jet} = v_{\rm jet} \cdot \tau_{\rm jet} \approx 6.3 \times 10^{15} \, \mathrm{m} \approx 0.2 \, \mathrm{pc}$$

3. Key Values

Quantity	Formula	Value
$P_{\rm orb}$	Spectroscopic	44 yr
$a_{\rm orb}$	Kepler 3rd law	~ 70 AU
$FUBi_i$	UQFF	-2.09×10^{-4} N
$M_{\rm total}$	Mira + WD	$\sim 2 M_{\odot}$
$v_{\rm jet}$	HST proper motion	~ 200 km/s
$t_{\rm jet}$	Jet age	$\sim 10^3$ yr

4. Physical Significance

R Aquarii provides UQFF calibration at the symbiotic nova (compact binary) scale intermediate between individual stellar objects (PAPER351 TDE) and galactic nuclei. The 44-year orbital period sets the longest UQFF binary activation period in the dataset (vs. PAPER347 Cen A 12.5-yr AGN cycle). The Kozima LENR coupling at the mass transfer interface raises the testable prediction that R Aquarii's nova outburst energetics deviate from standard accretion models by an LENR fraction, detectable in nuclear gamma-ray emission (e.g., INTEGRAL 511 keV line observations).

5. Deduplication Note

vs. PAPER_322 (R Aquarii in SOURCE122): Earlier treatment computed the 5-frequency resonances; this paper adds full $F_{\rm UBi_i}$ with $a_{\rm orb}$ from Kepler's third law and directly compares with HST 2025 jet data.

vs. PAPER_351 (ASASSN-14li): Different system class (symbiotic binary vs. TDE); both include $F_{\rm Kozima}$.

6. Classification

Physics Territory: FIRST UQFF FUBii for a symbiotic nova binary with Kepler aorb **Scale:** Stellar (70 AU binary, ~ 200 pc distance) **CP Implementation:** RAquariiSymbioticBinaryFUBiCalculator (CondensedPhysics3.py, Session 96)

UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = \frac{[SSq]GM/r}{5.0e-4 \times 0.57 \times 6.67e-11 M/r}$; for solar parameters: $U_{bi,Sun} = \frac{5.7e-4 \times 6.67e-11 \times 1.99e30}{(6.96e8)} = 1.47e+2$ m/s.

